Problem · Unit 1 · 10 points · 60 min · Standard exam level

P01 — Bead on a vertical helical wire: Frenet triad, constant angular acceleration and the reaction

A rigid wire has the shape of a circular helix of radius R wound around the vertical axis Oz of the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ (z upward, uniform gravity $-g\,\mathbf{k}$). In cylindrical coordinates (ρ, φ, z), with φ measured from the x axis and counter-clockwise seen from above, the wire is

$$\rho = R, \qquad z = b\,\varphi \qquad (b > 0)$$

so it rises a height $2\pi b$ (the pitch) per turn. A bead P of mass m slides ***without friction*** *along the wire. At $t = 0$ it is released from rest at $\varphi = \varphi_0$.*

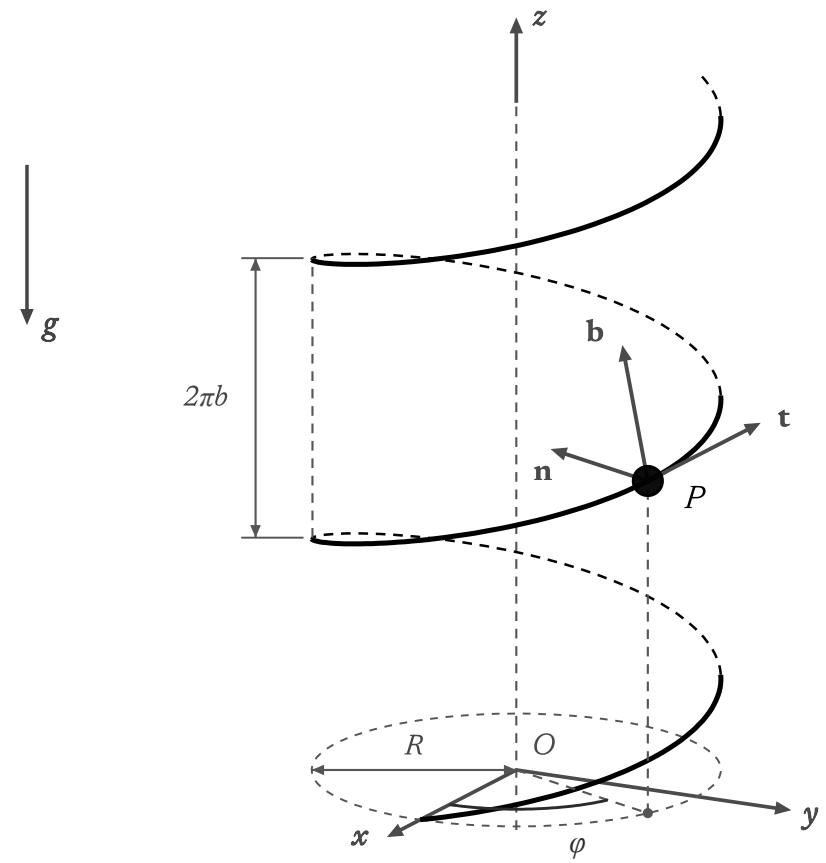


Figure 1. Bead P on the smooth helix $\rho = R$, $z = b\varphi$ (pitch $2\pi b$), z up and gravity $-g\,\mathbf{k}$; φ is the azimuth from the x axis, and $\{\mathbf{t}, \mathbf{n}, \mathbf{b}\}$ is the Frenet triad at P.

Use the cylindrical basis $\{\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{k}\}$ with $\mathbf{e}_\rho = \cos\varphi\,\mathbf{i} + \sin\varphi\,\mathbf{j}$, $\mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i} + \cos\varphi\,\mathbf{j}$, and the intrinsic (Frenet) triad $\{\mathbf{t}, \mathbf{n}, \mathbf{b}\}$ of the helix, with $\mathbf{t}$ oriented towards increasing φ. Write $h = \sqrt{R^2 + b^2}$.

Setup

- Data: m, R, b, g, φ_0. In the numerical part (f): $R = 0.30$ m, $b = 0.40$ m (so $h = 0.50$ m), $g = 9.81$ m s^{-2}.
- Hypotheses: point particle, rigid fixed wire, no friction (the reaction has no component along $\mathbf{t}$), S_0 inertial.
- The bead is threaded on the wire, so the constraint is **bilateral**: the reaction may point either way in the normal plane.
- Notation: the scalar b is the pitch parameter of the helix ($z = b\varphi$); the bold $\mathbf{b}$ is the binormal unit vector. They are unrelated.

Part (a) · 1.5 points

Degrees of freedom and kinematics

How many degrees of freedom has P? Classify the constraints. Write the position, velocity and acceleration of P in the cylindrical basis as functions of φ, $\dot\varphi$, $\ddot\varphi$.

Step 1. Count

The configuration of a free particle needs 3 coordinates (ρ, φ, z). The wire imposes $f_1 = \rho - R = 0$ and $f_2 = z - b\varphi = 0$: two independent holonomic equations (positions only), with no explicit time (scleronomic), and two-sided (bilateral). Hence $3 - 2 = 1$ degree of freedom (Notes §2.7), and φ is a convenient coordinate: ρ and z follow from it.

Step 2. Change of basis

The cylindrical basis turns with φ; its relation to the inertial basis is

$$\begin{pmatrix}\mathbf{e}_\rho & \mathbf{e}_\varphi & \mathbf{k}\end{pmatrix} = [R]\begin{pmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\end{pmatrix}, \qquad [R] = \begin{pmatrix}\cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1\end{pmatrix}$$

Checks: rows 1 and 2 have length $\cos^2\varphi + \sin^2\varphi = 1$ and $-\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$, row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \cos^2\varphi + \sin^2\varphi = +1$ (right-handed, $\mathbf{e}_\rho \times \mathbf{e}_\varphi = \mathbf{k}$). Differentiating the rows with the chain rule, $\dfrac{d\mathbf{e}_\rho}{d\varphi} = \mathbf{e}_\varphi$ and $\dfrac{d\mathbf{e}_\varphi}{d\varphi} = -\mathbf{e}_\rho$.

Step 3. Position and velocity

$\mathbf{r} = \rho\,\mathbf{e}_\rho + z\,\mathbf{k} = R\,\mathbf{e}_\rho + b\varphi\,\mathbf{k}$. The cylindrical velocity (Notes §2.6.2) is $\mathbf{v} = \dot\rho\,\mathbf{e}_\rho + \rho\dot\varphi\,\mathbf{e}_\varphi + \dot z\,\mathbf{k}$. Here $\dot\rho = 0$ and, by the chain rule, $\dot z = \dfrac{dz}{d\varphi}\dot\varphi = b\dot\varphi$:

$$\mathbf{v} = R\dot\varphi\,\mathbf{e}_\varphi + b\dot\varphi\,\mathbf{k}, \qquad |\mathbf{v}| = \sqrt{R^2\dot\varphi^2 + b^2\dot\varphi^2} = h\,|\dot\varphi|$$

Step 4. Acceleration

The cylindrical acceleration is $\mathbf{a} = (\ddot\rho - \rho\dot\varphi^2)\mathbf{e}_\rho + (\rho\ddot\varphi + 2\dot\rho\dot\varphi)\mathbf{e}_\varphi + \ddot z\,\mathbf{k}$. With $\rho = R$, $\dot\rho = \ddot\rho = 0$ and $\ddot z = \dfrac{d}{dt}(b\dot\varphi) = b\ddot\varphi$:

$$\mathbf{a} = -R\dot\varphi^2\,\mathbf{e}_\rho + R\ddot\varphi\,\mathbf{e}_\varphi + b\ddot\varphi\,\mathbf{k}$$

Check by direct differentiation of $\mathbf{v}$ with $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_\rho$: $\dot{\mathbf{v}} = R\ddot\varphi\,\mathbf{e}_\varphi + R\dot\varphi(-\dot\varphi\,\mathbf{e}_\rho) + b\ddot\varphi\,\mathbf{k}$, the same result.

KEY RESULTS

- 1 degree of freedom, coordinate φ; two holonomic, scleronomic, bilateral constraints $\rho - R = 0$, $z - b\varphi = 0$.
- $\mathbf{v} = R\dot\varphi\,\mathbf{e}_\varphi + b\dot\varphi\,\mathbf{k}$, $|\mathbf{v}| = h\,|\dot\varphi|$.
- $\mathbf{a} = -R\dot\varphi^2\,\mathbf{e}_\rho + R\ddot\varphi\,\mathbf{e}_\varphi + b\ddot\varphi\,\mathbf{k}$.

Part (b) · 2 points

Intrinsic triad, curvature and torsion

Obtain the unit vectors $\mathbf{t}$, $\mathbf{n}$, $\mathbf{b}$ *of the helix in the cylindrical basis, its curvature* κ *and torsion* τ, *and the intrinsic components of the acceleration of* P.

Step 1. Arc length and tangent

$ds = |\mathbf{v}|\,dt = h\,d\varphi$ (taking s increasing with φ), so $s = h(\varphi - \varphi_0)$ and $\dot s = h\dot\varphi$. The unit tangent (Notes §2.4) is

$$\mathbf{t} = \frac{d\mathbf{r}}{ds} = \frac{d\mathbf{r}}{d\varphi}\frac{d\varphi}{ds} = \frac{1}{h}\big(R\,\mathbf{e}_\varphi + b\,\mathbf{k}\big)$$

using $\dfrac{d\mathbf{e}_\rho}{d\varphi} = \mathbf{e}_\varphi$.

Step 2. Curvature and normal

With $\dfrac{d\mathbf{e}_\varphi}{d\varphi} = -\mathbf{e}_\rho$ and $\mathbf{k}$ constant:

$$\frac{d\mathbf{t}}{ds} = \frac{1}{h}\frac{d\mathbf{t}}{d\varphi} = \frac{1}{h}\cdot\frac{R}{h}\big(-\mathbf{e}_\rho\big) = \frac{R}{h^2}\,(-\mathbf{e}_\rho)$$

The modulus is the curvature and the unit vector the principal normal: $\kappa = R/h^2 = R/(R^2+b^2)$, $\mathbf{n} = -\mathbf{e}_\rho$. The normal is **horizontal** and points to the axis, although the helix is inclined.

Step 3. Binormal and torsion

$\mathbf{b} = \mathbf{t}\times\mathbf{n} = \dfrac{1}{h}(R\,\mathbf{e}_\varphi + b\,\mathbf{k})\times(-\mathbf{e}_\rho)$. With $\mathbf{e}_\varphi\times\mathbf{e}_\rho = -\mathbf{k}$ and $\mathbf{k}\times\mathbf{e}_\rho = \mathbf{e}_\varphi$:

$$\mathbf{b} = \frac{1}{h}\big(R\,\mathbf{k} - b\,\mathbf{e}_\varphi\big)$$

Then $\dfrac{d\mathbf{b}}{ds} = \dfrac{1}{h}\cdot\dfrac{-b}{h}\dfrac{d\mathbf{e}_\varphi}{d\varphi} = \dfrac{b}{h^2}\,\mathbf{e}_\rho = -\dfrac{b}{h^2}\,\mathbf{n}$, so $\tau = b/h^2 = b/(R^2+b^2)$. In matrix form the triad is $(\mathbf{t},\mathbf{n},\mathbf{b})^T = [Q]\,(\mathbf{e}_\rho,\mathbf{e}_\varphi,\mathbf{k})^T$ with rows $(0,\ R/h,\ b/h)$, $(-1,\ 0,\ 0)$, $(0,\ -b/h,\ R/h)$: the rows are unit and mutually orthogonal ($R^2/h^2 + b^2/h^2 = 1$, $-bR/h^2 + bR/h^2 = 0$), so $[Q][Q]^T = I$, and expanding along the second row $\det[Q] = -(-1)\big(\frac{R}{h}\frac{R}{h} + \frac{b}{h}\frac{b}{h}\big) = +1$: $\{\mathbf{t},\mathbf{n},\mathbf{b}\}$ is right-handed. The third Frenet formula $\dfrac{d\mathbf{n}}{ds} = -\kappa\,\mathbf{t} + \tau\,\mathbf{b}$ checks: $\dfrac{1}{h}\dfrac{d(-\mathbf{e}_\rho)}{d\varphi} = -\dfrac{\mathbf{e}_\varphi}{h}$ and $-\dfrac{R}{h^3}(R\mathbf{e}_\varphi + b\mathbf{k}) + \dfrac{b}{h^3}(R\mathbf{k} - b\mathbf{e}_\varphi) = -\dfrac{R^2+b^2}{h^3}\mathbf{e}_\varphi = -\dfrac{\mathbf{e}_\varphi}{h}$.

Step 4. Intrinsic acceleration

The intrinsic acceleration (Notes §2.5) is $\mathbf{a} = \ddot{s}\,\mathbf{t} + \kappa\dot{s}^2\,\mathbf{n}$. With $\dot{s} = h\dot{\varphi}$, $\ddot{s} = h\ddot{\varphi}$ and $\kappa\dot{s}^2 = \dfrac{R}{h^2}h^2\dot{\varphi}^2$:

$$\mathbf{a} = h\ddot{\varphi}\,\mathbf{t} + R\dot{\varphi}^2\,\mathbf{n}$$

Cross-check with (a): $\mathbf{a}\cdot\mathbf{t} = \dfrac{1}{h}(R\cdot R\ddot{\varphi} + b\cdot b\ddot{\varphi}) = h\ddot{\varphi}$, $\mathbf{a}\cdot\mathbf{n} = R\dot{\varphi}^2$, $\mathbf{a}\cdot\mathbf{b} = \dfrac{1}{h}(-bR\ddot{\varphi} + Rb\ddot{\varphi}) = 0$.

KEY RESULTS

- $\mathbf{t} = \dfrac{R\,\mathbf{e}_\varphi + b\,\mathbf{k}}{h}$, $\mathbf{n} = -\mathbf{e}_\rho$ (horizontal, towards the axis), $\mathbf{b} = \dfrac{-b\,\mathbf{e}_\varphi + R\,\mathbf{k}}{h}$.
- $\kappa = \dfrac{R}{R^2+b^2}$, $\tau = \dfrac{b}{R^2+b^2}$.
- $\mathbf{a} = h\ddot{\varphi}\,\mathbf{t} + R\dot{\varphi}^2\,\mathbf{n}$, no binormal component.

Part (c) · 1 point

Forces on the bead

Write the weight and the reaction of the wire in the intrinsic basis. How many unknowns does the problem have?

Step 1. Weight

$\mathbf{k}\cdot\mathbf{t} = b/h,\ \mathbf{k}\cdot\mathbf{n} = -\mathbf{k}\cdot\mathbf{e}_\rho = 0,\ \mathbf{k}\cdot\mathbf{b} = R/h$. Hence

$$\mathbf{W} = -mg\,\mathbf{k} = -\frac{mgb}{h}\,\mathbf{t} - \frac{mgR}{h}\,\mathbf{b}$$

The weight has no component along $\mathbf{n}$, because $\mathbf{n}$ is horizontal.

Step 2. Reaction

The wire is smooth: its reaction is orthogonal to the tangent, i.e. it lies in the normal plane spanned by $\mathbf{n}$ and $\mathbf{b}$ (Notes §6.5):

$$\mathbf{N} = N_n\,\mathbf{n} + N_b\,\mathbf{b}$$

The three scalar equations of Newton's law will determine the three unknowns $\varphi(t)$, N_n, N_b.

KEY RESULTS

- $\mathbf{k} = \dfrac{b}{h}\mathbf{t} + \dfrac{R}{h}\mathbf{b}$, so $\mathbf{W} = -mg\dfrac{b}{h}\,\mathbf{t} - mg\dfrac{R}{h}\,\mathbf{b}$.
- $\mathbf{N} = N_n\,\mathbf{n} + N_b\,\mathbf{b}$; unknowns $\varphi(t)$, N_n, N_b.

Part (d) · 2 points

Equation of motion and reaction

Obtain the equation of motion free of reactions and integrate it. Obtain N_n and N_b as functions of $\dot\varphi$. Does the bead move like a pendulum?

Step 1. Projections

With $\mathbf{a}$ of (b) and the forces of (c):

$$\mathbf{t}:\ mh\ddot{\varphi} = -\frac{mgb}{h}, \qquad \mathbf{n}:\ mR\dot{\varphi}^2 = N_n, \qquad \mathbf{b}:\ 0 = -\frac{mgR}{h} + N_b$$

Step 2. Equation of motion

The tangential equation contains no reaction:

$$\ddot{\varphi} = -\frac{gb}{h^2} = -\frac{gb}{R^2 + b^2}$$

The right-hand side does not depend on φ: there is **no restoring term** and no equilibrium. The tangential acceleration is $\ddot{s} = h\ddot{\varphi} = -g\,\dfrac{b}{h} = -g\sin\alpha$, with α the constant inclination of the helix ($\sin\alpha = b/h$): the bead moves exactly as on an inclined plane wound around a cylinder. It is not a pendulum.

Step 3. Integration

Integrating once between 0 and t (Barrow's rule, $\dot{\varphi}(0) = 0$): $\dot{\varphi}(t) - \dot{\varphi}(0) = \displaystyle\int_0^t \ddot{\varphi}\,dt' = -\frac{gb}{h^2}\,[t']_0^t$, so $\dot{\varphi} = -\dfrac{gb}{h^2}t$. Integrating again with $\varphi(0) = \varphi_0$:

$$\varphi(t) = \varphi_0 + \int_0^t \Big(-\frac{gb}{h^2}t'\Big)dt' = \varphi_0 - \frac{gb}{h^2}\Big[\frac{t'^2}{2}\Big]_0^t = \varphi_0 - \frac{gb}{2h^2}\,t^2$$

The bead descends (φ and $z = b\varphi$ decrease).

Step 4. Reactions

$$N_n = mR\dot{\varphi}^2, \qquad N_b = \frac{mgR}{h}$$

N_b supports the part of the weight normal to the wire and is constant; N_n (pointing to the axis) supplies the centripetal force of the horizontal circular motion and grows with the speed.

KEY RESULTS

- $\ddot{\varphi} = -\dfrac{gb}{R^2 + b^2}$: **constant**. $\varphi(t) = \varphi_0 - \dfrac{gb}{2h^2}t^2$.
- $N_n = mR\dot{\varphi}^2,\ N_b = \dfrac{mgR}{h}$ (constant).

Part (e) · 1.5 points

First integral

Show that the mechanical energy is conserved and use it to write $\dot{\varphi}^2$ and N_n as functions of φ.

Step 1. Power of the forces

$\mathbf{v} = h\dot{\varphi}\,\mathbf{t}$ and $\mathbf{N}\cdot\mathbf{t} = 0$, so $\mathbf{N}\cdot\mathbf{v} = 0$: the reaction of a fixed smooth curve does no work (Notes §6.4). The weight is conservative with $V = mgz = mgb\varphi$. Hence (Notes §4.5)

$$E = \tfrac{1}{2}m|\mathbf{v}|^2 + mgz = \tfrac{1}{2}mh^2\dot{\varphi}^2 + mgb\varphi = \text{const}$$

Step 2. Same result from the equation of motion

Multiply $h^2\ddot{\varphi} = -gb$ by $\dot{\varphi}$: $h^2\dot{\varphi}\ddot{\varphi} = \dfrac{d}{dt}\left(\tfrac{1}{2}h^2\dot{\varphi}^2\right)$ (chain rule) and $-gb\dot{\varphi} = \dfrac{d}{dt}(-gb\varphi)$. Integrating from 0 to t with Barrow's rule: $\tfrac{1}{2}h^2\dot{\varphi}^2 - 0 = -gb(\varphi - \varphi_0)$.

Step 3. Speed and reaction along the wire

$$\dot{\varphi}^2 = \frac{2gb(\varphi_0 - \varphi)}{h^2}, \qquad N_n = mR\dot{\varphi}^2 = \frac{2mgRb(\varphi_0 - \varphi)}{h^2}$$

With $\kappa = R/h^2$ and the height fallen $\Delta z = b(\varphi_0 - \varphi)$: $N_n = 2mg\kappa\,\Delta z$, i.e. $N_n = m\kappa v^2$ with $v^2 = 2g\Delta z$.

KEY RESULTS

- $E = \tfrac{1}{2}mh^2\dot{\varphi}^2 + mgb\varphi = mgb\varphi_0$.
- $\dot{\varphi}^2 = \dfrac{2gb(\varphi_0 - \varphi)}{h^2},\ N_n = \dfrac{2mgRb(\varphi_0 - \varphi)}{h^2} = 2mg\,\kappa\,\Delta z$ with $\Delta z = b(\varphi_0 - \varphi)$.

Part (f) · 2 points

One full turn: time, speed and reaction

Find the time t_1 the bead needs to complete one turn and its speed then. For $R = 0.30$ m, $b = 0.40$ m, $g = 9.81$ m s^{-2} give κ, τ, $\ddot{\varphi}$, t_1, the speed and the reaction (in units of mg) at t_1. Is the acceleration vector of the bead constant?

Step 1. Time for one turn

From $\varphi(t_1) = \varphi_0 - 2\pi$: $\dfrac{gb}{2h^2}t_1^2 = 2\pi \Rightarrow t_1 = \sqrt{\dfrac{4\pi h^2}{gb}} = 2h\sqrt{\dfrac{\pi}{gb}}$.

Step 2. Speed after one turn

From (e) with $\varphi_0 - \varphi = 2\pi$: $\dot{\varphi}_1^2 = \dfrac{4\pi gb}{h^2}$ and $v_1 = h|\dot{\varphi}_1| = \sqrt{4\pi gb}$. Equivalently $v_1 = h|\dot{\varphi}(t_1)| = h\dfrac{gb}{h^2}t_1 = \dfrac{gb}{h}\cdot 2h\sqrt{\dfrac{\pi}{gb}} = \sqrt{4\pi gb}$. Since the fall per turn is $2\pi b$, $v_1 = \sqrt{2g\cdot 2\pi b}$: the speed depends only on the height fallen, not on R.

Step 3. Numbers

$h = \sqrt{0.09 + 0.16} = 0.5$ m; $\kappa = 0.3/0.25 = 1.2$ m^{-1}; $\tau = 0.4/0.25 = 1.6$ m^{-1}; $\ddot{\varphi} = -9.81 \cdot 0.4/0.25 = -15.696$ rad s^{-2}, $\ddot{s} = h\ddot{\varphi} = -7.848$ m s^{-2} ($\sin\alpha = 0.8$, $\alpha = 53.13^\circ$).

$$t_1 = 2(0.5)\sqrt{\frac{\pi}{9.81\cdot 0.4}} = 0.8948\ \text{s}, \qquad v_1 = \sqrt{4\pi\cdot 9.81\cdot 0.4} = 7.022\ \text{m/s}, \qquad |\dot{\varphi}_1| = \frac{v_1}{h} = 14.04\ \text{rad/s}$$

Step 4. Reaction after one turn

$$\frac{N_n}{mg} = \frac{R\dot{\varphi}_1^2}{g} = \frac{4\pi Rb}{h^2} = 4\pi\cdot\frac{0.12}{0.25} = 1.92\pi \approx 6.032, \qquad \frac{N_b}{mg} = \frac{R}{h} = 0.6$$

$|\mathbf{N}| = mg\sqrt{6.032^2 + 0.6^2} \approx 6.062\,mg$. The tangential acceleration is constant, but $\mathbf{a} = h\ddot{\varphi}\,\mathbf{t} + R\dot{\varphi}^2\,\mathbf{n}$ has a growing normal part and a direction that turns with the triad: the acceleration vector is not constant.

KEY RESULTS

- $t_1 = 2h\sqrt{\dfrac{\pi}{gb}}$, $v_1 = \sqrt{4\pi g b} = \sqrt{2g\,(2\pi b)}$ (independent of R).
- $\kappa = 1.2$ m^{-1}, $\tau = 1.6$ m^{-1}, $\ddot{\varphi} = -15.70$ rad s^{-2}, $\ddot{s} = -7.848$ m s^{-2}, $t_1 = 0.8948$ s, $v_1 = 7.022$ m/s, $|\dot{\varphi}_1| = 14.04$ rad/s.
- $N_n = 1.92\pi\, mg \approx 6.032\, mg$, $N_b = 0.6\, mg$, $|\mathbf{N}| \approx 6.062\, mg$. The acceleration is not constant: its normal part $R\dot{\varphi}^2$ grows.

Interpretation and checks

- **Dimensional check:** κ and τ are m/m^2 = m^{-1}; gb/h^2 is (m s^{-2})(m)/m^2 = s^{-2}, an angular acceleration; $t_1 = 2h\sqrt{\pi/(gb)}$ has m/$\sqrt{\text{m}^2\,\text{s}^{-2}}$ = s; $N = mR\dot{\varphi}^2$ is kg m s^{-2} = N.
- **Limits:** $b \to 0$ (horizontal ring): $\ddot{\varphi} \to 0$, the bead stays at rest and $N_b \to mg$ carries the whole weight. $R \to 0$ (vertical straight wire): $\ddot{z} = b\ddot{\varphi} = -g\,b^2/h^2 \to -g$ (free fall), $\kappa \to 0$ and the reaction vanishes. κ is largest for $b \to 0$ and τ is largest for $b = R$.
- **Common mistakes in the exam:** (1) expecting a pendulum-type equation: on a helix the gravity component along the wire is the same everywhere, so $\ddot{\varphi}$ is constant; (2) taking $\mathbf{n}$ along the "radius of the coil" tilted with the helix: it is exactly $-\mathbf{e}_\rho$, horizontal; (3) forgetting N_b, which exists even at rest; (4) concluding that the acceleration is constant because $\ddot{\varphi}$ is: the normal acceleration $R\dot{\varphi}^2$ grows linearly with the height fallen.